
Intention-Level Alignment with Weak Supervision

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Abstract

We present *Self-Organizing Reinforcement Learning* (SoRL) for aligning LLMs under *weak supervision*. SoRL learns an internal, context-dependent objective via a Generative Abstraction Transformer that induces a hierarchical “inner language” whose time-varying abstractions guide token-level behavior. This yields (i) compositionality of goals, (ii) stability with reduced forgetting, and (iii) interpretability and efficiency by operating over compact abstractions instead of full chain-of-thought. We sketch a simple training loop and position SoRL against static alignment baselines, targeting gains in convergence and robustness when only sparse or noisy signals are available.

1 Motivation

As an analogy for LLM training, consider how humans learn. Humans rarely learn with direct supervision signals, such as a clear reward perfectly aligned with the ultimate goal or an optimal trajectory that guarantees success. Instead, we rely on weak supervision: partial, noisy, or indirect signals that only hint at progress. Take driving as an example. A learner driver never receives a precise scalar reward for “safe driving” while on the road. Instead, they infer it from scattered cues - avoiding honks, staying in lanes, following an instructor’s corrections, memories of rules in the driving manual. None of these signals fully specify the optimal trajectory, but together they shape a workable policy.

The same holds for training large language models. Rather than a single, unambiguous reward, models are guided by weak supervision signals: user preferences via implicit feedback, AI-generated labels, or correct execution of generated code. As with driving, a static reward function cannot capture the shifting composition of goals, intentions, and contexts that determine what “good” behavior looks like. Moreover, reward is an internal construct that often diverges from the external signals we provide (see for instance Karayanni and Nelken [2022] who highlight how organisms with identical external incentives can still develop different behaviors due to distinct internal reward systems).

$$\underbrace{r(s, a)}_{\text{External, Static Reward}} \rightarrow \underbrace{r_\theta(s, a \mid s^1)}_{\text{Internal, Dynamic Reward}}$$

Furthermore, unlike text classification, where Snorkel [Ratner et al., 2017] could aggregate discrete labeling functions into denoised training labels, generative AI must handle open-ended outputs where weak signals rarely reduce to a single class decision, making supervision both noisier and harder to reconcile. See Appendix A for more on the drawbacks of static reward functions and Appendix B for the history of weak supervision in training text classifiers.

2 Reframing reward

Developed at Thoughtworks, Self-Organizing Reinforcement Learning (SoRL) [Yu, 2025] addresses the challenge of turning noisy environmental signals into a stable, generalizable policy, where the notion of “good” behavior is often context-dependent.

SoRL replaces static external rewards with an internal, learnable function expressed through abstraction sequences that the model maintains. These abstractions act as latent symbols, forming an emergent inner-language that encodes shifting goals. They also provide scaffolds for implicit reasoning: abstractions can be chained or recomposed to capture subgoals and intermediate steps without explicit supervision.

Much like human language, which evolves to define new concepts (“quantum entanglement,” “climate justice”) or reinterpret existing ones (“cloud” in computing vs. weather), this inner-language allows models to reinterpret weak supervision signals in ways that remain consistent with situational demands Amir et al. [2025].

$$\underbrace{\pi_{\theta}(s)}_{\text{Language Model (Transformer)}} \rightarrow \underbrace{\pi_{\theta}(s \mid s^1, \dots, s^L)}_{\text{Generative Abstraction Transformer}}$$

Before we describe the specifics of how SoRL framework is implemented, we introduce 4 research questions around performance, interpretability and flexibility:

RQ1: Robustness and performance under weak supervision. Does SoRL achieve stronger performance than static methods (e.g., RLHF, RLAIF, reward-model distillation) on benchmarks where supervision is sparse or noisy, such as JailbreakBench [Chao et al., 2024], weakly-labeled QA datasets, or AI feedback generated alignment tasks? In particular, does its compositional abstraction mechanism improve generalization, reduce reliance on heuristics, and provide more stable convergence under weak signals?

RQ2: Emergence of explicit tokens for weak supervision signals. Do abstraction sequences in SoRL give rise to discrete, interpretable tokens that correspond to distinct weak supervision signals (e.g., correctness checks, preference feedback, or rejection signals), and can these tokens be consistently mapped across training runs or model scales?

RQ3: Detectability of weak signals via linear probes. Can linear probes trained on SoRL’s abstraction-level representations reliably identify which weak supervision signal (e.g., factuality vs. politeness vs. brevity) is being acted upon, and does probe accuracy improve relative to probing raw hidden states in baseline RLHF or RLAIF models?

RQ4: Circuit-level reconciliation of multiple signals. Do identifiable shorthands or circuits emerge in SoRL’s abstraction sequences that mediate conflicts or trade-offs between weak-signal tokens (e.g. balancing factuality vs. style)? Can activation patching as in Heimersheim and Nanda [2024] or causal scrubbing as in Geiger et al. [2025] be used to alter or override those circuits? For example, can we patch in the “correctness” token in a conflicting context to override a shallow heuristic, or conversely ablate a “shortcut” token to force reliance on a deeper abstraction?

Next we describe how the SoRL framework is implemented.

3 SoRL architecture.

Architecturally, where a standard language model learns a flat policy to predict the next token in natural language, SoRL introduces a Generative Abstraction Transformer (GAT) which maintains and updates multi-level abstraction sequences $s_l \in S^l$ for $l \leq L$, representing incrementally higher level abstract concepts. As illustrated in figure 1, higher level abstraction sequence are updated at less frequent timescales, therefore exhibiting more persistent effect and guides lower level executions. To generate an abstract token at level l , GAT conditions on current level context, as well as higher-level guidance and lower-level grounding, given

$$\pi_{\theta}(s^l \mid \underbrace{s^1, \dots, s^{l-1}}_{\text{lower-level grounding}}, \underbrace{s^{l+1}, \dots, s^L}_{\text{higher-level guidance}})$$

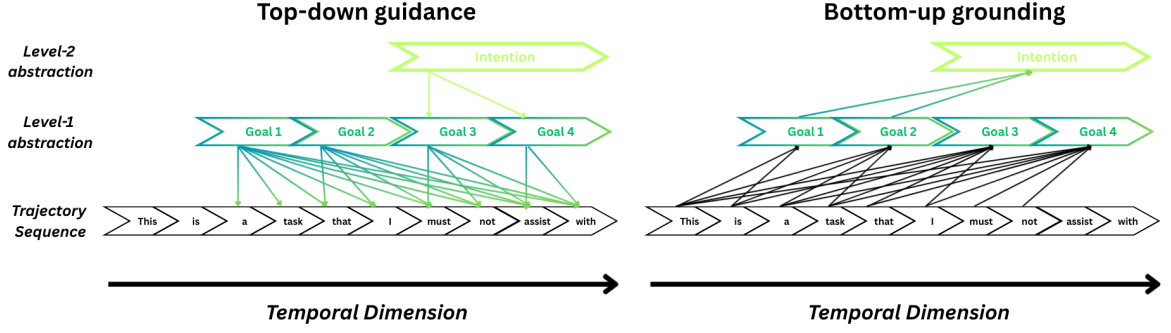


Figure 1: Generative Abstraction Transformer (GAT) with $L = 2$

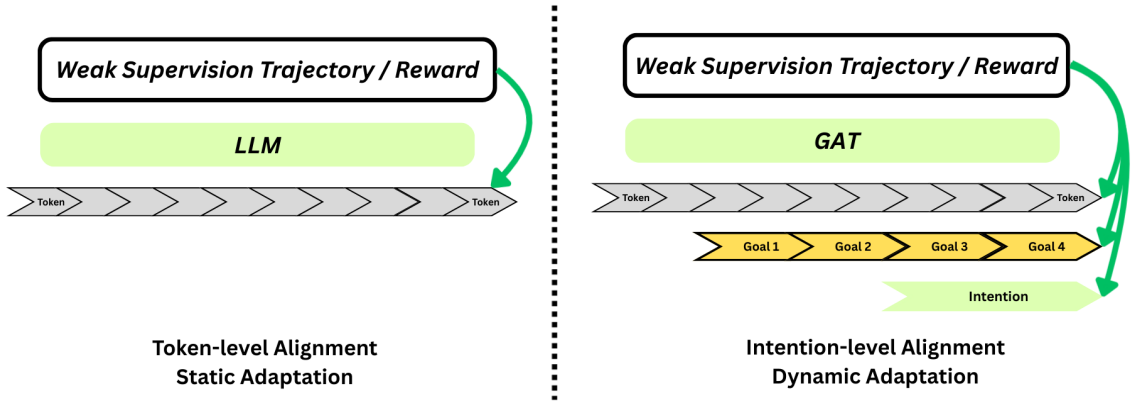


Figure 2: Intention-level Alignment with SoRL

Reward of abstraction at level l is determined by its effectiveness in improving execution at level $l - 1$, specifically we have

$$r_{\theta}(s^l | s^1, \dots, s^{l-1}, s^{l+1}, \dots, s^L) = \pi_{\theta}(s^1, \dots, s^{l-1} | s^l, \dots, s^L)$$

While weak supervision target or reward are directly assigned to trajectory sequence $s \in S$, we do not have supervision trajectory for abstract sequence, and therefore must learn from experience. We provide a pseudo algorithm for SoRL 1, which we will develop and test in the project.

Algorithm 1: SoRL

Input: Dataset $\mathcal{D} = \{s\}$ of trajectories, number of abstraction samples k , Generative Abstraction Transformer (GAT) with parameters θ

while not converged **do**

 Sample minibatch $S \subset \mathcal{D}$;

foreach $s \in S$ **do**

 // Step 1: Sample candidate abstractions

 Sample k abstractions $\{s^{1,i}\}_{i=1}^k \sim \pi(\cdot | s)$;

 // Step 2: Select best abstraction

$s^{1,*} \leftarrow \arg \max_{i \leq k} \pi(s^{1,i} | s)$;

 // Update GAT

$\mathcal{L}(\theta) \leftarrow -\log \pi(s, s^{1,*})$;

 Update θ with one gradient step on $\mathcal{L}(\theta)$;

4 Experimental setup

Our experiments will focus on aligning open-source foundation models such as Qwen and LLaMA. We evaluate across diverse benchmarks, including JailbreakBench, where supervision is naturally weak due to AI-generated labels, retrieval-oriented tasks such as TriviaQA, Natural Questions, HotpotQA, and WebQuestions, as well as general evaluation suites such as Arena-Hard, MMLU, AIME, and Codeforces that cover reasoning, knowledge, and coding ability. The central objective is to compare SoRL’s adaptive alignment with static alignment methods such as Supervised Fine-Tuning (SFT) and Reward-augmented Fine-Tuning (RLFT) when reward signals are available. Beyond benchmarking, we are equally interested in an interpretability study of the abstract tokens learned by the Generative Abstraction Transformer (GAT), examining what semantic or functional roles these tokens represent and how intervening on them can be used to control the model’s downstream behavior.

Weak Type	Supervision	Description	Benchmarks / Signals	Typical Heuristic Failures (Observed/Mismatched)	SoRL Advantage (Hyp.)
1. Noisy / AI-Labels		Rewards from imperfect annotators or model graders.	<i>JailbreakBench</i> , AI feedback safety/alignment tasks	Overfitting to grader quirks; “refusal keyword” gaming; prompt-style mimicry instead of true policy change.	Abstraction tokens separate safety/compliance signals from grader noise; recombination improves transfer across prompts.
2. Sparse Correctness Checks		Only pass/fail at end of solution/program.	<i>AIME</i> , <i>Codeforces</i> , strict final-answer math/code eval	Template regurgitation; brittle pattern-matching; shallow CoT padding without solving; length correctness.	Abstractions carry subgoal structure (such as function signatures), interpolating between sparse checks to reduce template reliance.
3. Preference Feedback		Human/AI pairwise rankings over outputs.	<i>Arena-Hard</i> , RLHF/RLAIF preference pairs	Sycophancy; verbosity/fluency outranking factuality; politeness over safety; reward hacking of style.	Tokens disentangle style vs. truth/safety; targeted adaptation balances competing signals without global drift.
4. Proxy / Heuristic Rewards		Rule-based or correlational proxies for success.	Retrieval QA: <i>NQ</i> , <i>HotpotQA</i> , <i>WebQuestions</i> (lexical/embedding proxies)	Keyword matching; lexical overlap; embedding-similarity maximization; shallow passage copying.	Circuits reconcile “proxy” tokens with “correctness” tokens; improves multi-hop reasoning and out-of-domain transfer.

Table 1: Four weak-supervision regimes, aligned failure modes, and SoRL’s hypothesized advantages. Heuristics like verbosity, keyword matching, and embedding similarity appear where they are known to mislead training objectives.

5 Benefits

We expect the following benefits for solving weak supervision:

- **Compositionality:** SoRL learns to compose simple weak abstractions into complex “sentences” that predict and optimize for sparse, decision-level goals.
- **Interpretability and Efficiency:** SoRL can be viewed as RL over a compressed Chain-of-Thought, allowing us to interpret the model’s intentions without the massive overhead of aligning full CoT traces or the ambiguity of trying to decode massive activation spaces. So for instance, if certain tokens of our evolved language correspond to a weak supervision signal, we can detect easier if the model overfits poorly to one of the weak signals of supervision.
- **Dynamic Adaptation:** SoRL-trained models adapt their behavior to context by generating different inner-language “sentences,” enabling flexible adjustment without retraining.

Crucially, SoRL has a theoretical backbone. We have shown in our prior work in Yu [2025] that while classical RL must relearn policies from scratch when rewards change, SoRL can probably locate the optimal goal-oriented policy by composing previously learned policies. To our knowledge, this is the first RL algorithm to demonstrate such compositional optimality.

Looking ahead, we envision a SoRL Adapter (SLM): a lightweight, LLM-agnostic module that learns this inner language, making robust, interpretable alignment a plug-and-play component for any future model.

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A The problems of static reward

Current predominant methods for LLM alignment often fall into one of two categories, both implicitly or explicitly fitting a static reward function. Methods like Reinforcement Learning from Human Feedback (RLHF) Ouyang et al. [2022] collect human preferences to train a reward model, which then provides a fixed proxy reward signal. Reinforcement Learning with Verifiable Reward (RLVR) DeepSeek [2024], Zheng et al. [2025], Yu et al. [2025] relies on a pre-programmed reward function. Constitutional AI Bai et al. [2022] and Reinforcement Learning from AI Feedback (RLAIF) Lee et al. [2024], Yuan et al. [2025], Zhao et al. [2025] use LLM’s to provide scores or feedback, which again are distilled into a static reward proxy.

While these methods have advanced LLM capabilities, their reliance on a static reward function—whether in a reward model, a verification program, or an LLM score—fundamentally clashes with the dynamic nature of human intent and weak supervision. Disadvantages include:

- Static adaptation methods for RL risk catastrophic forgetting Luo et al. [2025].
- Models also fall back on superficial heuristics - for example, RLAIF can reward verbosity over relevance.
- With RLHF, models inherit cultural biases from labelers and can become overly compliant to unsafe requests if framed politely.
- Moreover, alignment is frequently shallow Qi et al. [2024], shifting only the opening tokens of a response while leaving downstream behavior unstable.

Recently, online reinforcement learning approaches, such as those used in models like Cursor Tab tab [2025], have been adopted to constantly update models based on user acceptance rate and has shown significant improvements in performance. This demonstrates the critical importance of dynamic alignment. However, online training and updating new model checkpoint introduces significant cost and overhead.

B Weak Supervision in Text Classification: Lessons from Snorkel

A major line of work on weak supervision predates LLMs and focused on classical text classification. The Snorkel framework demonstrated that instead of relying on scarce, hand-labeled data, models can be trained from a set of labeling functions: heuristics, rules, distant supervision, or external knowledge bases that provide noisy but overlapping signals of class membership. By modeling the accuracies and correlations of these weak signals, Snorkel learns a denoised probabilistic label that can supervise downstream classifiers Ratner et al. [2017, 2020].

For example, in sentiment analysis, one labeling function might assign “positive” if the text contains words like *great* or *excellent*, while another might rely on emojis or exclamation marks. Individually unreliable, these signals can be combined to generate higher-quality training labels. This paradigm was shown to match or surpass the performance of models trained on smaller, fully labeled datasets while reducing annotation costs.